specialized equipment and trained crews. These methods increased operational costs and deployment times, especially in combat zones.

- *Maintenance and reliability issues*: UAV systems of this generation require frequent calibration, component replacement, and post-flight inspection. Limited durability of engines, control surfaces, and electronics led to high maintenance-to-flight-hour ratios.
- *Weather sensitivity*: Second-generation UAVs lacked advanced flight control algorithms for turbulence correction or wind compensation. Operations were often grounded due to rain, fog, or strong crosswinds, affecting mission continuity.
- *Limited real-time data processing*: Although telemetry enabled realtime video transmission, data processing capabilities were minimal. Most image interpretation and analysis occurred post-flight, delaying decision-making in time-sensitive missions.
- *Training and operational complexity*: The lack of automation meant that skilled operators and technicians were required for every flight operation, making UAV missions resource-intensive and limiting large-scale deployment.

3.6.3 Strategic and regulatory limitations

- *Restricted civilian integration*: The use of drones outside the military was minimal due to regulatory uncertainty and airspace restrictions. Civil aviation authorities lacked frameworks for UAV certification, airworthiness, and safety compliance.
- *High production and operational costs*: The cost of developing and maintaining drones such as the *RQ-2 Pioneer* or *CL-289* was substantial. These systems required sophisticated materials, propulsion